Optimizing Inventory and Demand Forecasting for Spare Parts

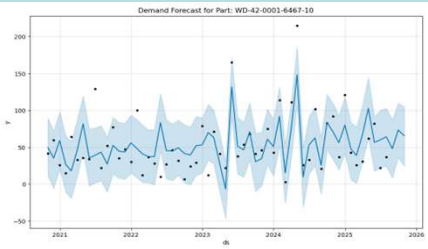
MSc DS CST4090

Presenter: Snehal D Souza
Supervisor: Dr. Ivan Reznikov

A41

INTRODUCTION

Inventory forecasting plays a critical role in inventory management. Manual based approaches often lead to delays and other inefficiencies. This project develops an automated, cloud-based demand forecasting pipeline utilizing AWS technologies and the Prophet model, to forecast demand and classify stock based on their monthly movement.



Aim:

"Develop a cloud-based demand forecasting and spare parts classification system using time-series modeling to enhance inventory efficiency and decision-making."

Why is accurate demand forecasting important?

- *Customer Satisfaction*
- *Operational Excellence*
- *Cost Efficiency*

Current Challenges faced in inventory forecasting

Reliance on Manual Based Approaches
Forecasts are frequently generated using spreadsheets and static reports, which are time-consuming and prone to human error.

Disconnected Systems
Many organizations lack seamless connectivity between their ERP systems and modern analytics tools, resulting in fragmented data flows.

Need for Scalable, Cloud-Based Solutions
Organizations require flexible, automated pipelines capable of processing large datasets in real time to improve accuracy and responsiveness.

METHODOLOGY

A structured approach was followed to develop and deploy the pipeline:

Data Extraction from the ERP System
- Inventory Ledger data exported in csv format
- Data includes part numbers, transaction quantities and stock balances
- All modern ERP's have the option to export data.

Data Upload to AWS S3
- 3-Folder structure: Input, Output & Historic
- Files are uploaded directly to the input folder
- Provides secure, scalable and low-latency storage.
- StarPoint of automatic data processing pipeline.

AWS Lambda Function
- Acts as the automation trigger for new uploads.
- Reads environment variables, container path, etc.
- Launches SageMaker processing Job with specified instance size and configuration.
- Trigger is through S3 Event Notification.

AWS ECR and Docker Environment
- A custom docker image was built locally and deployed to the Elastic Container Registry (ECR).
- Included required dependencies: Prophet, Pandas, boto3, etc.
- Ensured consistent running environment and reduced run-time.

SageMaker Processing Job
- Executes the forecasting script using Docker.
- Cleans and merges data, runs Prophet for time-series forecasting.
- Classification algorithm classifies stock based on movement.
- Ensures consistent automation and reliability.

AWS QuickSight Visualization
- Connected to data using JSON manifest files.
- Interactive dashboards display forecast trends, inventory gaps and movement classes.
- Auto-refresh enables new processing outputs to be reflected in the dashboard in real time.

Prophet model used to forecast 3-month demand per part, leveraging historical issue trends. It handles seasonality, missing data, and irregular patterns in time-series.

CLASSIFICATION LOGIC

Based on the latest transaction date and forecast behaviour:
- **Fast Moving:** Regular moving.
- **Slow Moving:** Irregular movement
- **Dead Stock:** No issues for 12+ months

Extra fields were added for more in-depth classification.

RESULTS

The inferences from the pipeline were visualized using **AWS QuickSight**, providing clear and interactive **real-time** dashboards for **monitoring** inventory trends, demand forecasts, and stock movement classifications."

One Click Solution:
Seamless end-to-end data upload, processing, forecasting, and output through AWS services.

Intelligent Forecasting & Classification:
Prophet model accurately predicted 3-month demand and categorized parts by movement (fast, slow, or dead stock).

Interactive Visualization:
Real-time dashboards in QuickSight enabled faster, data-driven inventory decisions.

CONCLUSIONS

- Demonstrates how ARIMA and Prophet models can be used to generate clear, actionable inventory forecasts.
- Few studies use cloud services for end-to-end forecasting. AWS enables automation, scalability, and real-time decision support.
- Enables managers to anticipate stock shortages, identify slow or dead inventory, optimize procurement planning.
- Manual forecasting is prone to bias and inconsistency. Data-driven approaches use historical trends to deliver accurate, actionable insights.

FUTURE WORK

- *RAG based Chat Bot*
- *Deeper models like LSTM, XGBoost*
- *Integration with existing ERP's*

REFERENCES

1. Hyndman, R.J. & Athanasopoulos, G. (2018),
2. Amazon Web Services (2015).
3. Docker. (2025).
4. Meta Open Source. (2025).